

Лабораторная работа №12

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2025

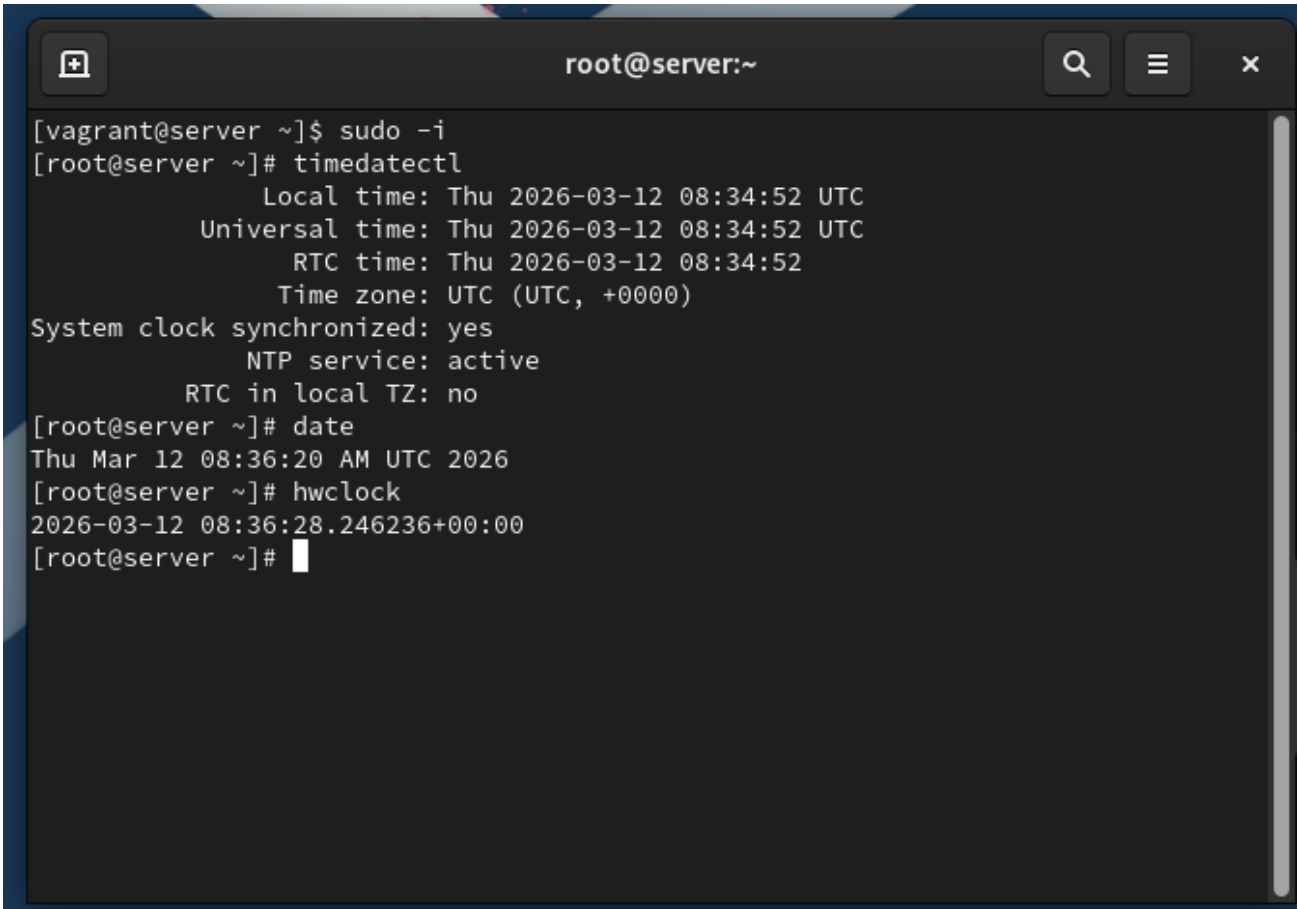
Лабораторная работа №12

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На сервере посмотрим параметры настройки даты и времени, текущего системного времени и аппаратного времени (рис. @fig-1):

A screenshot of a terminal window titled 'root@server:~'. The terminal shows a user switching from 'vagrant' to 'root' using 'sudo -i'. Then, the 'timedatectl' command is executed, displaying system time details: Local time (Thu 2026-03-12 08:34:52 UTC), Universal time (Thu 2026-03-12 08:34:52 UTC), RTC time (Thu 2026-03-12 08:34:52), and Time zone (UTC (UTC, +0000)). It also shows 'System clock synchronized: yes', 'NTP service: active', and 'RTC in local TZ: no'. Next, the 'date' command is run, showing 'Thu Mar 12 08:36:20 AM UTC 2026'. Finally, the 'hwclock' command is executed, showing '2026-03-12 08:36:28.246236+00:00'.

```
root@server:~  
[vagrant@server ~]$ sudo -i  
[root@server ~]# timedatectl  
        Local time: Thu 2026-03-12 08:34:52 UTC  
        Universal time: Thu 2026-03-12 08:34:52 UTC  
        RTC time: Thu 2026-03-12 08:34:52  
        Time zone: UTC (UTC, +0000)  
System clock synchronized: yes  
        NTP service: active  
        RTC in local TZ: no  
[root@server ~]# date  
Thu Mar 12 08:36:20 AM UTC 2026  
[root@server ~]# hwclock  
2026-03-12 08:36:28.246236+00:00  
[root@server ~]#
```

Рис. 1: Этап 1

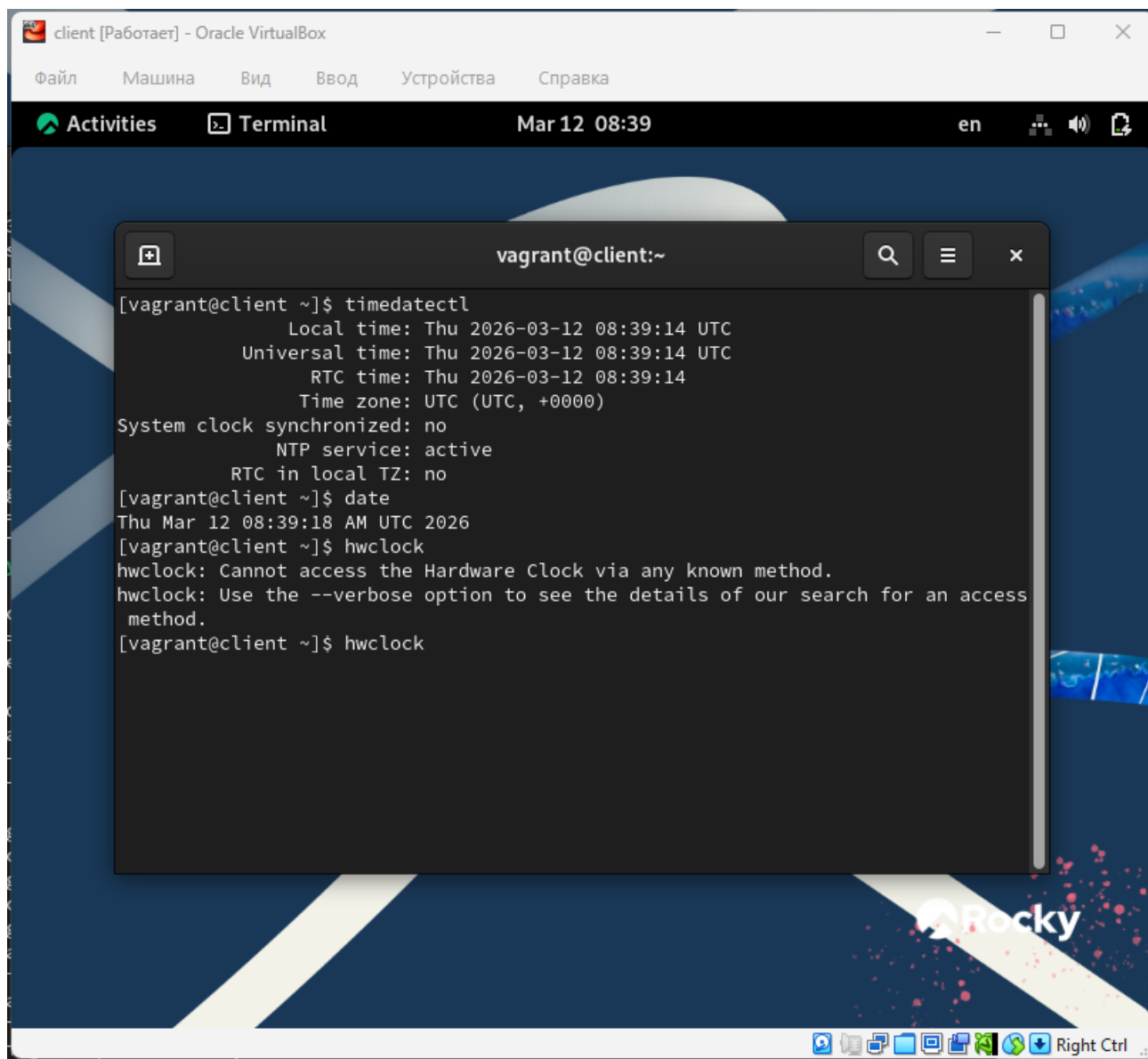
```
root@server:~
2026-03-12 08:36:28.246236+00:00
[root@server ~]# dnf -y install chrony
Rocky Linux 9 - BaseOS      8.9 kB/s | 4.3 kB    00:00
Rocky Linux 9 - BaseOS      5.7 MB/s | 16 MB     00:02
Rocky Linux 9 - AppStream    12 kB/s | 4.8 kB     00:00
Rocky Linux 9 - AppStream    6.9 MB/s | 17 MB     00:02
Rocky Linux 9 - Extras       5.9 kB/s | 3.1 kB     00:00
Rocky Linux 9 - Extras      18 kB/s | 17 kB      00:00
Package chrony-4.5-1.el9.x86_64 is already installed.
Dependencies resolved.
=====
Package           Architecture Version           Repository      Size
=====
Upgrading:
chrony             x86_64        4.6.1-2.el9      baseos          340 k
Transaction Summary
=====
Upgrade 1 Package

Total download size: 340 k
Downloading Packages:
chrony-4.6.1-2.el9.x86_64.rpm      3.5 MB/s | 340 kB    00:00
-----
Total                               800 kB/s | 340 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Running scriptlet: chrony-4.6.1-2.el9.x86_64  1/2
  Upgrading      : chrony-4.6.1-2.el9.x86_64    1/2
  Running scriptlet: chrony-4.6.1-2.el9.x86_64  1/2
  Running scriptlet: chrony-4.5-1.el9.x86_64    2/2
  Cleanup       : chrony-4.5-1.el9.x86_64       2/2
```

Рис. 2: Этап 2

На клиенте посмотрим параметры настройки даты и времени, текущего системного времени и аппаратного времени (рис. @fig-2):

Установим на сервере необходимое программное обеспечение chrony (рис. @fig-3):



```
client [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка
Activities  Terminal  Mar 12 08:39  en

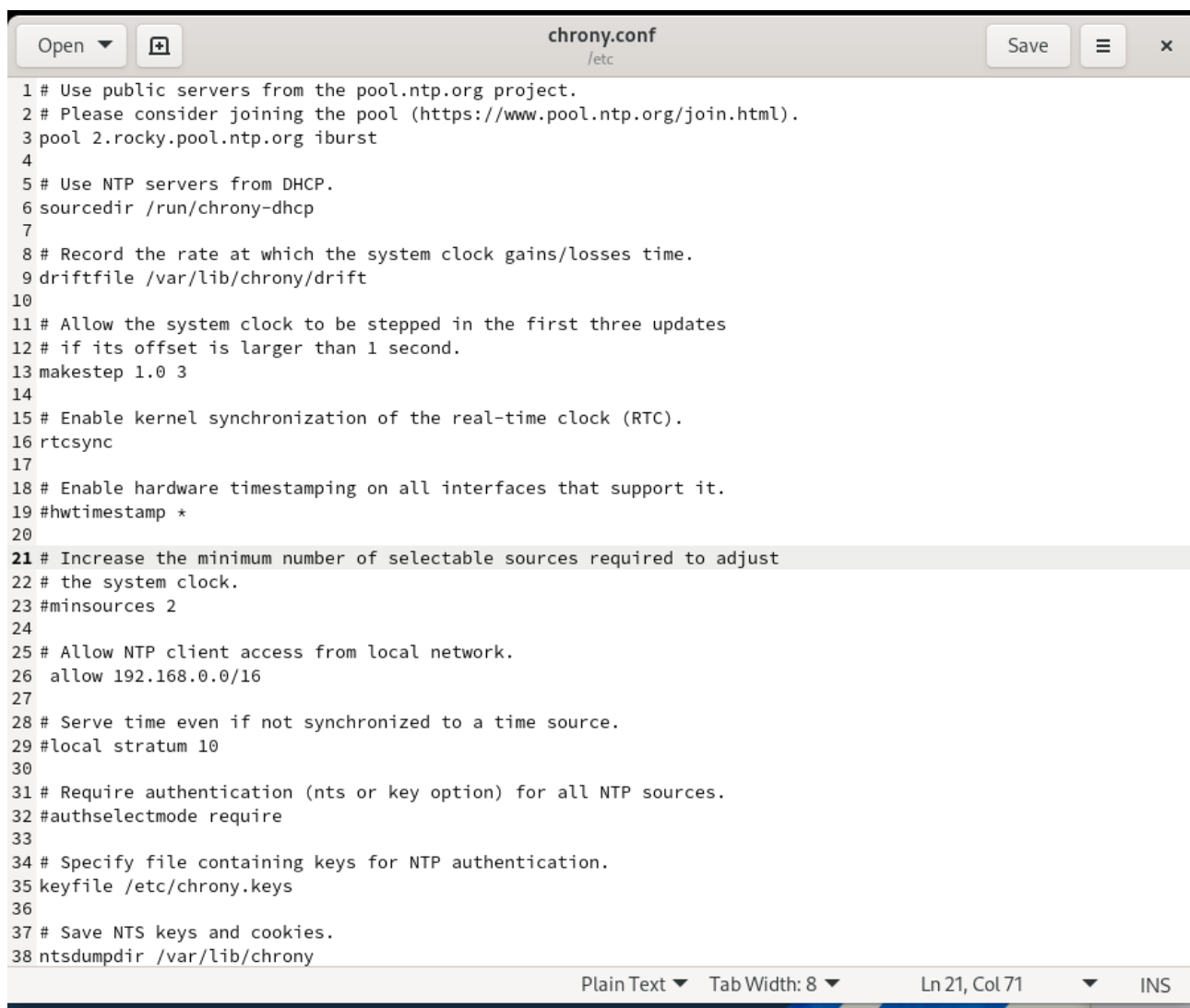
[vagrant@client ~]$ timedatectl
          Local time: Thu 2026-03-12 08:39:14 UTC
          Universal time: Thu 2026-03-12 08:39:14 UTC
            RTC time: Thu 2026-03-12 08:39:14
            Time zone: UTC (UTC, +0000)
System clock synchronized: no
          NTP service: active
          RTC in local TZ: no
[vagrant@client ~]$ date
Thu Mar 12 08:39:18 AM UTC 2026
[vagrant@client ~]$ hwclock
hwclock: Cannot access the Hardware Clock via any known method.
hwclock: Use the --verbose option to see the details of our search for an access
method.
[vagrant@client ~]$ hwclock
```

Рис. 3: Этап 3

Проверим источники времени на клиенте до настройки (рис. @fig-4):

Проверим источники времени на сервере до настройки (рис. @fig-5):

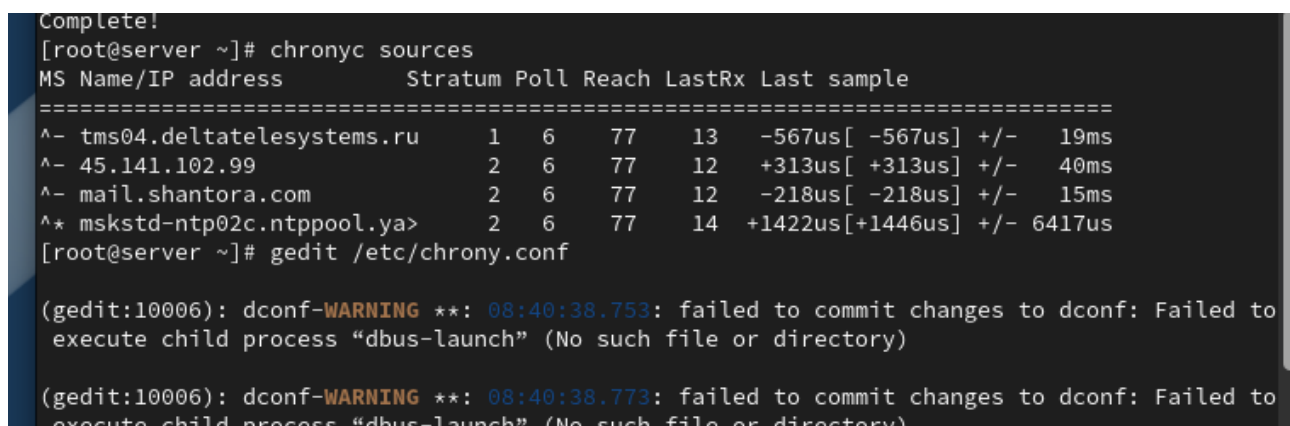
На сервере откроем на редактирование файл `/etc/chrony.conf` и добавим строку `allow 192.168.0.0/16` (рис. @fig-6):



```
1 # Use public servers from the pool.ntp.org project.
2 # Please consider joining the pool (https://www.pool.ntp.org/join.html).
3 pool 2.rocky.pool.ntp.org iburst
4
5 # Use NTP servers from DHCP.
6 sourcedir /run/chrony-dhcp
7
8 # Record the rate at which the system clock gains/losses time.
9 driftfile /var/lib/chrony/drift
10
11 # Allow the system clock to be stepped in the first three updates
12 # if its offset is larger than 1 second.
13 makestep 1.0 3
14
15 # Enable kernel synchronization of the real-time clock (RTC).
16 rtsync
17
18 # Enable hardware timestamping on all interfaces that support it.
19 #hwtimestamp *
20
21 # Increase the minimum number of selectable sources required to adjust
22 # the system clock.
23 #minsources 2
24
25 # Allow NTP client access from local network.
26 allow 192.168.0.0/16
27
28 # Serve time even if not synchronized to a time source.
29 #local stratum 10
30
31 # Require authentication (nts or key option) for all NTP sources.
32 #authselectmode require
33
34 # Specify file containing keys for NTP authentication.
35 keyfile /etc/chrony.keys
36
37 # Save NTS keys and cookies.
38 ntsdumpdir /var/lib/chrony
```

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Рис. 4: Этап 4

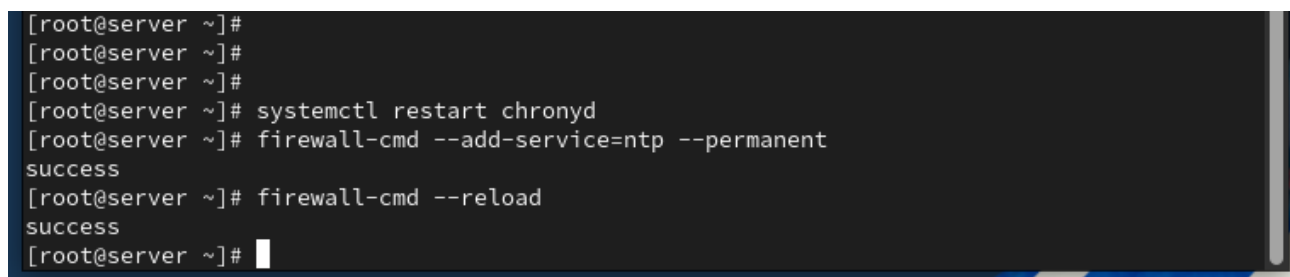


```
Complete!
[root@server ~]# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^-- tms04.deltatelesystems.ru    1    6    77    13   -567us[ -567us] +/-  19ms
^-- 45.141.102.99                2    6    77    12   +313us[ +313us] +/-  40ms
^-- mail.shantora.com            2    6    77    12   -218us[ -218us] +/-  15ms
^* mskstd-ntp02c.ntppool.ya>    2    6    77    14  +1422us[+1446us] +/- 6417us
[root@server ~]# gedit /etc/chrony.conf

(gedit:10006): dconf-WARNING **: 08:40:38.753: failed to commit changes to dconf: Failed to
execute child process "dbus-launch" (No such file or directory)

(gedit:10006): dconf-WARNING **: 08:40:38.773: failed to commit changes to dconf: Failed to
execute child process "dbus-launch" (No such file or directory)
```

Рис. 5: Этап 5



```
[root@server ~]#
[root@server ~]#
[root@server ~]#
[root@server ~]# systemctl restart chronyd
[root@server ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]#
```

Рис. 6: Этап 6

На сервере перезапустим службу chronyd и настроим межсетевой экран (рис. @fig-7):

```
1 # Use public servers from the pool.ntp.org project.
2 # Please consider joining the pool (https://www.pool.ntp.org/join.html).
3 pool 2.rocky.pool.ntp.org iburst
4
5 # Use NTP servers from DHCP.
6 sourcedir /run/chrony-dhcp
7
8 server server.vagrant.net iburst
9
10 # Record the rate at which the system clock gains/losses time.
11 driftfile /var/lib/chrony/drift
12
13 # Allow the system clock to be stepped in the first three updates
14 # if its offset is larger than 1 second.
15 makestep 1.0 3
16
17 # Enable kernel synchronization of the real-time clock (RTC).
18 rtsync
19
20 # Enable hardware timestamping on all interfaces that support it.
21 #hwtimestamp *
22
23 # Increase the minimum number of selectable sources required to adjust
24 # the system clock.
25 #minsources 2
26
27 # Allow NTP client access from local network.
28 #allow 192.168.0.0/16
29
30 # Serve time even if not synchronized to a time source.
31 #local stratum 10
```

Рис. 7: Этап 7

На клиенте откроем файл /etc/chrony.conf и добавим строку server server.user.net iburst, удалив все остальные строки с директивой server (рис. @fig-8):

```
[vagrant@client ~]$ systemctl restart chronyd
Failed to restart chronyd.service: Connection timed out
See system logs and 'systemctl status chronyd.service' for details.
[vagrant@client ~]$ chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^? 51.250.35.68              0      6      0      -      +0ns[  +0ns] +/-    0ns
^? 77.50.202.25              0      6      0      -      +0ns[  +0ns] +/-    0ns
^? 91.188.214.68             0      6      0      -      +0ns[  +0ns] +/-    0ns
^? 92.255.126.1              0      6      0      -      +0ns[  +0ns] +/-    0ns
[vagrant@client ~]$
```

Рис. 8: Этап 8

На клиенте перезапустим службу chronyd (рис. @fig-9):

Проверим источники времени на клиенте после настройки (рис. @fig-10):

```
[root@server ~]# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^* 6.18.241.92.s-inform.net    3   6   377   50  -1080us[ -973us] +/-  12ms
^- 89.188.118.150             3   6   377   51  +1639us[+1747us] +/- 108ms
^+ time.cloudflare.com        3   6   377   51  +3099us[+3206us] +/-  16ms
^+ hcvvv7-real.sci-nnov.ru    2   6   377   51   -932us[ -825us] +/-  21ms
[root@server ~]#
```

Рис. 9: Этап 9

```
[root@server ~]#
[root@server ~]# systemctl restart chronyd
[root@server ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^* 6.18.241.92.s-inform.net    3   6   377   50  -1080us[ -973us] +/-  12ms
^- 89.188.118.150             3   6   377   51  +1639us[+1747us] +/- 108ms
^+ time.cloudflare.com        3   6   377   51  +3099us[+3206us] +/-  16ms
^+ hcvvv7-real.sci-nnov.ru    2   6   377   51   -932us[ -825us] +/-  21ms
[root@server ~]#
```

Рис. 10: Этап 10

Проверим источники времени на сервере после настройки (рис. @fig-11):

```
[root@server ~]# systemctl restart chronyd
[root@server ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^* 6.18.241.92.s-inform.net    3   6   377   50  -1080us[ -973us] +/-  12ms
^- 89.188.118.150             3   6   377   51  +1639us[+1747us] +/- 108ms
^+ time.cloudflare.com        3   6   377   51  +3099us[+3206us] +/-  16ms
^+ hcvvv7-real.sci-nnov.ru    2   6   377   51   -932us[ -825us] +/-  21ms
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/ntp/etc
[root@server server]# cp -R /etc/chrony.conf /vagrant/provision/server/ntp/etc/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch ntp.sh
[root@server server]# chmod +x ntp.sh
[root@server server]# gedit ntp.sh
(gedit:10069): dconf-WARNING **: 08:49:36.462: failed to commit changes to dconf: Failed to
```

Рис. 11: Этап 11

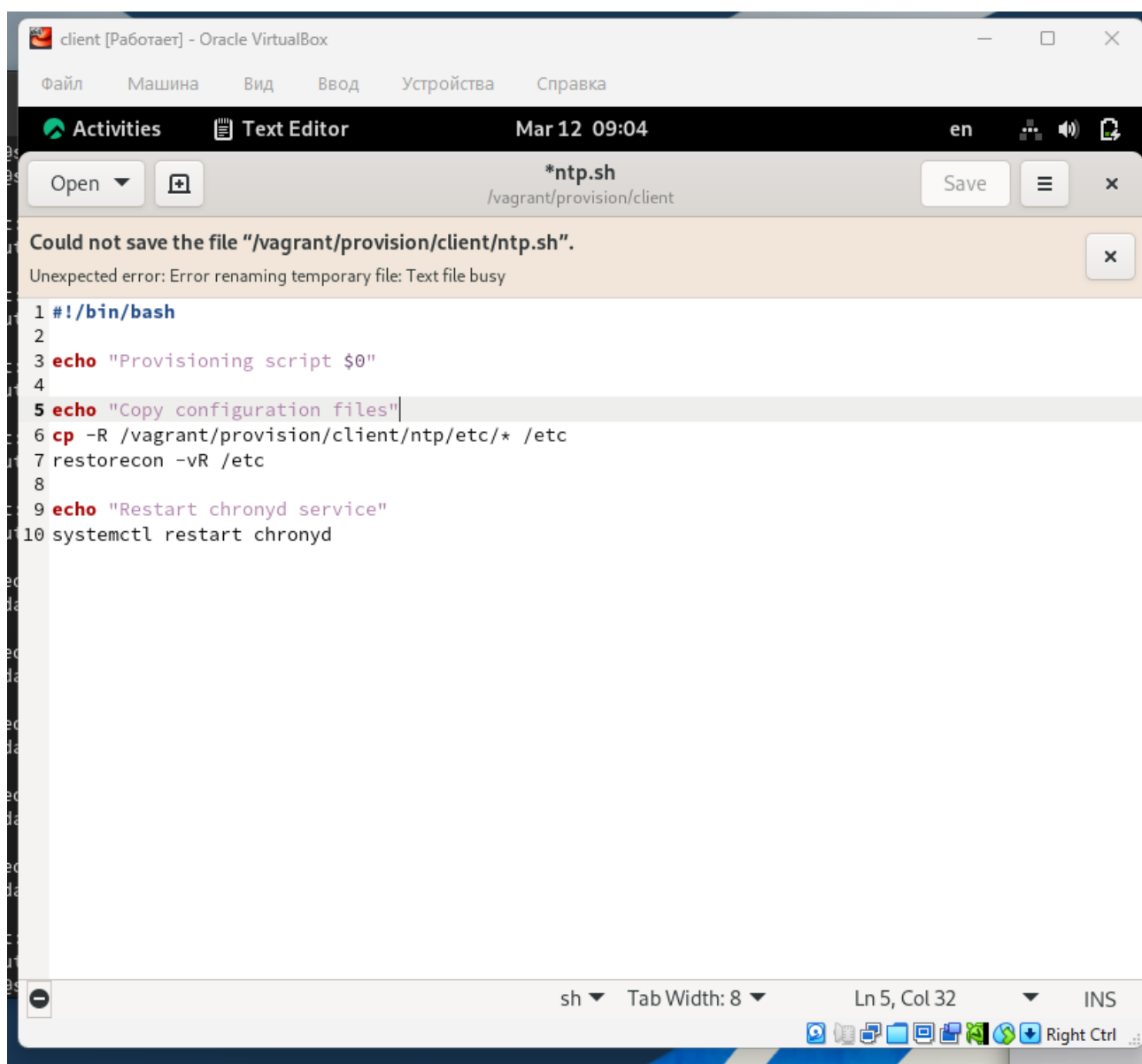
На виртуальной машине server перейдём в каталог для внесения изменений в настройки внутреннего окружения, создадим каталог ntp и исполняемый файл ntp.sh (рис. @fig-12):

Откроем файл ntp.sh на редактирование и пропишем скрипт из лабораторной работы (рис. @fig-13):



```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install chrony
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/server/ntp/etc/* /etc
7 restorecon -vR /etc
8 echo "Configure firewall"
9 firewall-cmd --add-service=ntp
10 firewall-cmd --add-service=ntp --permanent
11 echo "Restart chronyd service"
12 systemctl restart chronyd
```

Рис. 12: Этап 12



```
Could not save the file "/vagrant/provision/client/ntp.sh".
Unexpected error: Error renaming temporary file: Text file busy

1 #!/bin/bash
2
3 echo "Provisioning script $0"
4
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/client/ntp/etc/* /etc
7 restorecon -vR /etc
8
9 echo "Restart chronyd service"
10 systemctl restart chronyd
```

Рис. 13: Этап 13

На виртуальной машине client перейдём в каталог для внесения изменений в настройки внутреннего окружения, создадим каталог ntp и исполняемый файл ntp.sh (рис. @fig-14):

```
91     run: always ,
92     path: "provision/client/01-routing.sh"
93   client.vm.provision "client ntp",
94     type: "shell",
95     preserve_order: true,
96     path: "provision/client/ntp.sh"
97
98   client.vm.provider :virtualbox do |v|
99     v.linked_clone = true
100   # Customize the amount of memory on the VM
101     v.memory = 4096
```

Рис. 14: Этап 14

Откроем файл ntp.sh на редактирование и пропишем скрипт из лабораторной работы (рис. @fig-15):

```
47   server.vm.provision "server ntp",
48     type: "shell",
49     preserve_order: true,
50     path: "provision/server/ntp.sh"
51   server.vm.provider :virtualbox do |v|
52     v.linked_clone = true
53   # Customize the amount of memory on the VM
54     v.memory = 4096
```

Рис. 15: Этап 15

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?